Unsupervised Concept Discovery for Medical Vision–Language Models:
A Rigorous Characterization of MedConcept Limits to a 2D BiomedCLIP Adaptation

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ABSTRACT

Medical Vision–Language Models (VLMs) are accurate but opaque. MedConcept [?] targets *3D volumetric* abdominal CT with the Merlin [?] foundation model—a resource-intensive setting. We *adapt* its paradigm—sparse-autoencoder (SAE) decomposition of a frozen VLM's embeddings, alignment to a radiology vocabulary, LLM-as-judge evaluation—to a *2D*, GPU-accessible setting on BiomedCLIP [?], at two scales: IU X-Ray [?] (∼7k) and ROCOv2 [?] (∼80k, 10×). This deliberate accessibility trade yields compressed 2D embeddings. We first expose the slot-wise cross-seed Jaccard stability headline as *floor-by-construction* (a relabelled single SAE scores ≈0.0077) and replace it with a permutation-invariant *matched* metric against a subspace-conditioned null. Under it, the 2D adaptation is genuinely non-identifiable: best-match cosine 0.30–0.33 at only 1.67–1.81× the null, ≈0% matched at ≥0.9, naming 0.40–0.48; neither a 10× corpus nor a pre-projection variant lifts this, localising the constraint to the representation site—the price of decomposing a 2D compressed, contrastive-shaped global vector. SPLiCE [?], a deterministic fixed-dictionary decomposition, is the sole judge-ready method (81.6% Aligned, MedGemma-4B vs. 3.3%, Llama-3.1-8B). We deliver a rigorous, partially-negative characterisation of an accessible 2D adaptation—consistent with MedConcept's own report of median Aligned< 0.1 on one of its datasets.

KEYWORDS

Concept-based explainability, sparse autoencoders, medical vision–language models, interpretability evaluation, reproducibility

1 INTRODUCTION

Medical VLMs attain strong performance on pathology classification, segmentation, and report generation, yet their internal representations are high-dimensional, polysemic, and opaque—a serious concern in safety-critical clinical use that classical post-hoc tools (saliency, attention) rarely resolve into semantic structure.

Concept-based explainability reframes an explanation in terms of human-interpretable units. Where supervised approaches such as Concept Bottleneck Models [?] and TCAV [?] require a pre-defined, labelled concept set, recent work attempts *unsupervised* discovery: decomposing a frozen model's representations into sparse features, naming each feature against a clinical vocabulary, and validating the result quantitatively. MedConcept [?] instantiates this as a three-stage pipeline (sparse-activation extraction, vocabulary alignment, LLM-judge evaluation) with the Aligned/Unaligned/Uncertain metrics—but targets *3D volumetric* abdominal CT via the Merlin [?] foundation model, a compute-heavy setting out of reach for course-level hardware.

We make a deliberate, accessibility-driven choice: *adapt* the MedConcept paradigm to a 2D setting built on BiomedCLIP [?] and 2D radiographs, which runs on a single consumer GPU. The trade-off is explicit—2D projected embeddings are cheaper to extract but far more compressed than Merlin's volumetric features—so we treat this as a study of *where and why* the accessible 2D adaptation succeeds or falls short, attributing any shortfall to the representation rather than to a training bug. We ask whether the concepts such a pipeline discovers are *reproducible* and *clinically valid*, and find—rigorously, against analytical nulls—that the learned 2D sparse decomposition is essentially non-identifiable: independently seeded SAEs share no canonical features, the naming signal is weak, and many features never fire. Crucially, we first expose that the slot-wise index Jaccard used as a stability headline is *floor-by-construction* (a relabelled copy of one SAE scores ≈0.0077 on it), so non-identifiability must be measured permutation-invariantly.

We probe three mitigations (a 512-d **baseline** SAE; **Path A**, a 768-d pre-projection variant; **Path B**, SPLiCE's fixed dictionary) plus a concept-**organisation** extension, and contribute: a *2D, GPU-accessible adaptation* of MedConcept with an analysis of why it underperforms the 3D/Merlin setting; a *relabeling control* showing the slot-wise Jaccard is floor-by-construction; a permutation-invariant *matched-stability* metric with a subspace-conditioned null under which the concepts are non-identifiable; a 10× scale refutation; and an honest, partially-negative evaluation.

2 RELATED WORK

Concept-based explainability. Concept Bottleneck Models [?] restructure a network to predict an explicit concept layer before the classifier, trading minor accuracy for auditable, concept-level intervention. TCAV [?] tests directional sensitivity of a logit to user-defined Concept Activation Vectors. Both presuppose supervised, pre-defined concepts—precisely the limitation unsupervised discovery targets.

Sparse autoencoders for monosemantic decomposition. Bricken et al. [?] showed that overcomplete dictionary-learning autoencoders trained on transformer activations yield individual latents that activate on coherent, human-nameable features. Gao et al. [?] scaled the Top-K (k-sparse) autoencoder, which sets the L_0 sparsity exactly via top-k gating and removes the L1 sparsity penalty, improving the reconstruction–sparsity frontier; we adopt this architecture.

SAE stability and universality. Lan et al. [?] showed SAE feature spaces across models are similar only up to rotation (a shared subspace, not identical features), and Leask et al. [?] argued SAEs "do not find canonical units of analysis." This weak-vs-strong universality motivates our matched-stability metric: we expect only a shared subspace, not identical features.

Sparse concept decomposition. SPLiCE [?] bypasses a learned dictionary and decomposes a frozen CLIP embedding directly into a sparse, non-negative combination over a fixed, interpretable concept dictionary. Because the dictionary is fixed and the method deterministic, it sidesteps the non-identifiability of learned factorisation—a property we exploit as a data-frugal baseline.

Medical VLMs and the reference framework. BiomedCLIP [?] is a domain-specific contrastive VLM (ViT-B/16 + PubMedBERT, PMC-15M) whose image encoder exposes a 768-d pre-projection hidden state; its checkpoint projects this to the 512-d shared space we decompose. MedConcept [?] provides the template (sparse extraction, vocabulary alignment, LLM-judge) we adapt; it operates on *3D volumetric* abdominal CT with the Merlin [?] foundation model (far more compute- and data-demanding than our 2D setting). RadLex [?] and MeSH [?] the external radiology lexicons for naming.

3 RESEARCH GAPS

From the literature and the project brief we identify five gaps that this work addresses:

(1) **Instability / non-identifiability:** sparse factorisations are not reproducible across seeds; no prior medical-VLM work reports seed-stability against a null.
(2) **Clinical validity:** a cosine-assigned name is cosmetic unless it tracks real pathology, yet concepts are rarely validated against ground truth.
(3) **Representation-location mismatch:** the SAE literature operates on raw hidden states, but reference pipelines fit the SAE on the already-projected CLIP space.
(4) **Small-data robustness:** learned SAEs need 10^5–10^6 activations; medical corpora are 2–3 orders smaller, yielding dead features.
(5) **Non-reproducible evaluation:** single-metric claims never report a chance floor (slot-wise overlap is degenerate under per-seed permutation); metrics must be null-calibrated and permutation-invariant.

4 METHODOLOGY

4.1 Backbone, vocabulary, and split

We use BiomedCLIP (ViT-B/16, 768-d hidden state projected by frozen visual_projection : $\mathbb{R}^{768} \to \mathbb{R}^{512}$ into shared contrastive space; images 224×224, L2-normalised). The shared 512-d space exhibits a *modality gap* [?]; we estimate it as the difference of the mean image and vocabulary embeddings and subtract it before cosine naming/decomposition.

Vocabulary (external, eval-independent). Because naming must not be circular with the judge's ground truth, we use two *external* lexicons chosen to match each corpus's domain: **RadLex** [?], filtered to chest terms (39 anchor queries across 13 sub-domains, +14 NIH ChestX-ray14 seeds, ∼1,030 terms), for the chest X-ray domain of IU X-Ray; and **MeSH** [?] (NLM; radiology branches A/C/E, ∼1,024 terms) for ROCOv2's broader multimodal radiology, where a chest-specific lexicon is inappropriate. Both are built by encoding terms with the text encoder and ranking by cosine to the anchors; both are free and licence-free (a deliberate alternative to the licence-gated UMLS the reference uses).

Corpora and split. IU X-Ray [?] (7,470 frontal/lateral images across 3,852 radiographic studies) is the primary, small-scale setting; ROCOv2 (∼80k images) is a 10×-scale replication that tests the data-volume hypothesis directly. To avoid leakage we split IU X-Ray *by radiographic study* (key = `patient_IM-study`), so no study appears in both partitions; ROCOv2 is split by image (no shared patient structure). The IU split yields 5,955 train / 1,515 test images (3,081 / 771 studies) with verified group overlap = 0, recomputed deterministically on every run (sorted study set, fixed seed).

4.2 Baseline: Top-K SAE on the 512-d projected space

We train a Top-K SAE [?] ($\mathrm{ReLU}(W_{\mathrm{enc}}(x - b_{\mathrm{dec}}) + b_{\mathrm{enc}})$, hard top-$k$ sparsification, b_{dec} at the geometric median) on the 512-d image embeddings. Configuration: D=2048, k=32, 8,000 steps, lr=5×10^{-5}, batch size 256, five seeds $(0, 42, 123, 456, 789)$. Concept naming assigns each *live* (non-dead) decoder row to its closest vocabulary term by cosine similarity, after gap correction ($W_{\mathrm{dec}} \leftarrow W_{\mathrm{dec}} - \mathrm{gap}$). Per-image pseudo-reports list the top-5 active concepts with a stated dominant concept.

4.3 Path A: Top-K SAE on the 768-d hidden state

Path A addresses Gap 3. Instead of visual_projection$(\cdot)$, we extract the 768-d CLS token *before* the projection and train an identical Top-K SAE on these raw (un-normalised) hidden states. Because the text vocabulary lives in 512-d, we bridge the two spaces with the *frozen* projection: $\hat{W}^{512}_{\mathrm{dec}} = W^{768}_{\mathrm{dec}} W^{\top}_{\mathrm{proj}}$, gap-correct, then name as in the baseline. Dead features are detected pre-projection and cross-checked with an activation-based mask.

4.4 Path B: SPLiCE on a fixed dictionary

Path B addresses Gaps 1 and 4. We decompose each gap-corrected 512-d image embedding into a sparse, non-negative combination over the fixed vocabulary dictionary with a two-stage solver (SPLiCE-style, in place of its original Lasso): Orthogonal Matching Pursuit selects k=32 atoms, then non-negative least-squares re-solves the coefficients on that support. The dictionary is never estimated from the samples, so the method is deterministic by construction and has no cross-seed instability. The output schema mirrors the SAE explanations so the same judge can score it.

4.5 Concept-organisation extension

Addressing Gap 5, we cluster the active vocabulary into concept families by agglomerative clustering (average linkage, cosine metric). Each cluster is annotated with the most specific shared anatomical ancestor, with two degeneracy guards: leaf-root rejection and a *subtree-size canopy* that rejects ancestors whose descendant count exceeds 1% of the ontology (e.g. the trivially generic "anatomical

Table 1: Baseline Top-K SAE (D=2048, k=32); IU X-Ray (RadLex) vs. ROCOv2 (MeSH, 10×). "Sub./iso. null" = observed best-match over the subspace-conditioned/isotropic null.

	IU X-Ray (1×)	ROCOv2 (10×)
Reconstruction cosine	0.991	0.968
Dead features (full test)	15.0–17.8%	0.5–0.8%
Matched cosine (obs)	0.299	0.327
Decoder erank	363	357
Subspace null ratio (honest)	1.67×	1.81×
Isotropic null ratio (lower bnd)	1.98×	2.16×
Frac. matched ≥ 0.9	0.0%	0.3%
Naming cosine (live, mean)	0.40	0.48
Slot-wise Jaccard (deprecated)	0.0077	0.0077

entity" node). The stage emits structured per-image families plus organisation metrics (silhouette, redundancy reduction, coverage); it is standalone and does not feed the judge.

4.6 Evaluation metrics

Stability (matched, the headline). For each decoder row in seed i we take the best cosine match across all rows of seed j [**? ?**], a permutation-invariant statistic. The null is a *subspace-conditioned* random-vector draw: real decoder directions concentrate in a subspace of effective rank erank=$(\Sigma\sigma)^2/\Sigma\sigma^2$≈357–363 (not the ambient 512), so we draw null vectors inside each decoder's top-erank right-singular subspace and compare against the other decoder projected into it; the isotropic (full-512-d) ratio is a reported lower bound. **Stability (slot-wise, deprecated).** Cross-seed Jaccard of top-k index sets is *floor-by-construction*—not invariant to per-seed feature permutation (Sec. 5.2); retained only as a deprecated identical-permutation check. **Naming.** Mean cosine of the assigned term, plus the fraction matched at ≥0.7 and ≥0.9 (random directions cannot reach 0.9). **Faithfulness.** Point-biserial correlation of feature activations with in-distribution IU X-Ray MeSH/Problems labels, with a per-feature shuffle null. **LLM judge.** Each concept is scored Aligned/Unaligned/Uncertain against the reference report by a prompted LLM (MedGemma-4B [**?**] and, for cross-LLM comparison, Llama-3.1-8B); we report the raw %Aligned as a measure of *local* plausibility.

5 EXPERIMENTS AND ANALYSIS

5.1 The 2D baseline is a non-identifiable failure case (Gap 1)

Table 1 reports the baseline on both corpora. Reconstruction is faithful (cosine 0.968–0.991) yet cross-seed feature identity is absent: matched cosine 0.30–0.33 at only 1.67–1.81× the null, ≤0.3% matched at ≥0.9, naming capped at 0.40–0.48 (0% above 0.7). This is genuine non-identifiability (init distinct, decoders full-rank, no collapse); the deprecated slot-wise Jaccard lands at its ≈0.0077 floor (Sec. 5.2).

Table 2: Baseline (512-d) vs. Path A (768-d), parity config.; "obs/null" over the isotropic null.

	Baseline 512-d	Path A 768-d
Slot-wise Jaccard	0.0084	0.0083
Analytical floor	0.0079	0.0079
Dead features (full test)	16.0%	1.7%
Naming cosine (live)	0.42	0.47
Matched cosine (obs)	0.299	0.325
Matched cosine (null)	0.151	0.124
obs/null	1.97×	2.63×
Frac. matched ≥ 0.9	≈0%	≈0%

Table 3: Path A dictionary/k sweep (768-d, 8k steps).

Preset	D	k	Dead	Matched obs/null
Conservative	1024	16	41.7%	2.78×
Default	2048	32	12.9%	2.63×
Aggressive	4096	64	6.6%	2.00×

5.2 Relabeling control: the slot-wise metric is floor-by-construction

The slot-wise Jaccard cannot distinguish "two different SAEs" from "one SAE whose columns were renamed." Permuting one trained ROCOv2 SAE's rows by a random bijection π (function-preserving; reconstruction unchanged to 10^{-4}) makes slot-wise Jaccard collapse to 0.0077 (indistinguishable from real cross-seed) while the matched metric correctly reports identity (1.0, frac≥0.9=1.0). The slot-wise signal is pure permutation noise—a binary identical-permutation check, not a stability metric. All later stability claims use the matched metric.

5.3 Neither representation depth nor data scale restores identifiability

Table 2 contrasts Path A (768-d pre-projection hidden state) with the baseline. Operating on the 768-d hidden state markedly reduces dead features (1.7% vs. 16%) and improves naming cosine (0.47 vs. 0.42). Yet the permutation-invariant matched metric is unchanged in its verdict: both share a decoder subspace above the null (isotropic 1.97×/2.63×, collapsing to ≈1.3–1.5× under the subspace-conditioned null), and almost no features match strongly (0% at ≥0.9)—*weak universality* [**? ?**]: a shared subspace exists, but not canonical, reproducible features. Path A still operates on the *pooled CLS token*, so the representation-site mismatch is only partially addressed.

We then test data scale (Gap 4) directly. Training the baseline on ROCOv2 (~80k, 10× over IU) improves training health (dead features 16%→0.6%) but leaves identifiability flat (Table 1): matched cosine 0.299→0.327, subspace ratio 1.67×→1.81×, frac≥0.9 0.0%→0.3%. A 10× corpus lifts none of these above the non-identifiability regime.

A dictionary/k sweep on Path A (Table 3) confirms the pattern: matched obs/null falls from 2.78× (D=1024) through 2.63× (default) to 2.00× (D=4096) as overcompleteness grows—no setting escapes weak universality.

Table 4: Concept organisation across sources.

Source	Active	Clusters	Silhouette	Empty img. (/1515)
SPLiCE	997	32	0.020	0
SAE baseline	80	9	0.094	0
SAE hidden	14	4	0.284	1260

5.4 Why the 2D adaptation underperforms the 3D MedConcept setting

This negative identifiability result is specific to the accessible 2D adaptation, not to the MedConcept paradigm: MedConcept decomposes Merlin's high-dimensional *3D volumetric* features [?], whereas we decompose BiomedCLIP's 2D *projected* global vector. We attribute the gap to three structural factors: *(i)* Merlin is a larger 3D CT foundation model than the lighter 2D BiomedCLIP. This difference leads to a post-projection 512-d contrastive vector that has effective rank erank≈357–363, so a 2048-way dictionary is overcomplete and the factorisation non-identifiable, whereas Merlin's volumetric embedding is wider and higher-rank; *(ii)* the SAE literature targets raw residual streams, not a pooled, L2-normalised, contrastive-shaped summary—Path A (pre-projection CLS) only partially relieves this; lastly, *(iii)* learned SAEs need 10^5–10^6 activations, and even our 10× corpus stays 1–2 orders below. The binding constraint is thus the *2D compressed representation* we adopted for GPU accessibility; the experiment our diagnosis predicts would restore identifiability is an SAE on the BiomedCLIP *patch-token residual stream* (mid layer, not pooled) [?].

5.5 Path B is deterministic and judge-ready

SPLiCE decomposes every test image with a fixed, randomness-free solver (OMP selects the k=32-atom support, NNLS re-solves coefficients), so the result is deterministic by construction with no cross-seed instability, and its SAE-compatible schema lets the same judge read SPLiCE concepts. On IU X-Ray it covers all 1,515 test images in 9.5 s using 997/1,031 RadLex terms (96.7%, 14.18 concepts/image); on ROCOv2 (~80k) it covers all 15,958 images in 101 s using 1,024/1,024 MeSH terms (100%, 8.56 concepts/image). Gap correction raises the maximum atom–image cosine sharply on both corpora (IU 0.49⟶0.85, ROCOv2 0.54⟶0.86), confirming the gap, not the solver, was binding; frequent terms are mixed (a property of the supplied dictionaries, not of SPLiCE).

5.6 Concept organisation is dense but weakly separated

Applied to all three IU sources (Table 4), SPLiCE groups 997 concepts into 32 families, cutting concepts-per-image 14.18⟶8.09 (1.75×) but with weak silhouette (0.020); the noisy IU vocabulary makes every cluster inherit a noisy ancestor. The rebuilt SAE baseline exposes 80 active concepts (9 families, silhouette 0.094, no empty images) that inherit the baseline's non-identifiability, and Path A's hidden source stays degenerate (14 concepts, 1,260/1,515 empty). ROCOv2 SPLiCE replicates non-degenerately (1,024 concepts, 32 families, silhouette 0.066): density scales with corpus size, separation stays weak.

5.7 A small upper tail of features is genuinely faithful

Despite global instability, a faithfulness ablation (D=2048 baseline, 5 seeds) finds 17.8% ± 0.9 of live features exceed the shuffle-null 95th percentile in point-biserial correlation with IU X-Ray MeSH/Problems labels (1.1% at $|r| > 0.30$; strongest $|r|$=0.459±0.057 on a *mass* feature)—a real but small upper tail atop an otherwise degenerate dictionary: concepts are simultaneously *unstable cross-seed* (zero cluster across seeds at ≥0.90, not above a seed-tag shuffle-null, p=0.17) and *partially faithful* (Gap 2).

5.8 The judge finds high but model-dependent local plausibility

Running the LLM-judge to convergence (temperature=0, zero parse errors) with two backbones—MedGemma-4B [?] and Llama-3.1-8B—SPLiCE on IU X-Ray scores 81.6% Aligned under MedGemma (1,230/1,508) but only 3.3% under Llama; the degenerate Path A hidden source scores 76.3% vs. 0.5%; the SAE baseline on ROCOv2 88.3% (14,095/15,957) vs. 23.1% (Llama leans *Uncertain*). This *local* plausibility is orthogonal to the global identifiability verdict and is model-dependent. Low alignment is a property of the paradigm itself: MedConcept reports median Aligned< 0.1 on AbdomenAtlas [?]—so our scores are consistent with the harshness of report-conditioned concept verification (given the different backbone and modality), not evidence the 2D setting is uniquely broken.

6 CONCLUSIONS

We adapted MedConcept to a 2D, GPU-accessible setting (BiomedCLIP + 2D radiographs). Because the slot-wise stability headline is floor-by-construction (a relabelled SAE scores ≈0.0077), we measure non-identifiability permutation-invariantly: under a matched metric with a subspace-conditioned null, the 2D SAE concepts are non-identifiable (≈0% at ≥0.9, naming 0.40–0.48), and neither Path A nor a 10× corpus lifts it—a structural limit of decomposing a 2D compressed, contrastive-shaped global vector, not a training bug (Sec. 5.4). SPLiCE on a fixed dictionary is the sole deterministic, judge-ready alternative, and a small (~18%) upper tail of features is genuinely faithful.

Limitations and future work. (i) The LLM-judge returns *model-dependent* verdicts, orthogonal to identifiability and not directly comparable to MedConcept's (different backbone/modality/judge); random-k null explanations score comparably (79.6% MedGemma, 2.9% Llama), indicating weak discriminability. (ii) The predicted mitigation is an SAE on the BiomedCLIP *patch-token residual stream* (mid layer, not pooled CLS) [?], testing whether the 2D limit is intrinsic to the representation site. (iii) Our 10× scale test (ROCOv2) also broadens the *domain* (multimodal, not just chest), confounding scale with breadth; the clean test is PadChest (~160k chest X-rays, same IU X-Ray domain), begun but unfinished—to test whether scale alone restores reproducibility.

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